

2.4 Exact Equations

Introduction Although the simple differential equation $ydx + xdy = 0$ is separable, we can solve it in an alternative manner by recognizing that the left-hand side is equivalent to the differential of the product of x and y ; that is, $ydx + xdy = d(xy)$. By integrating both sides of the equation we immediately obtain the implicit solution $xy = c$.

Differential of a Function of Two Variables If $z = f(x, y)$ is a function of two variables with continuous first partial derivatives in a region R of the xy -plane, then its **differential** (also called the total differential) is

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy. \quad (1)$$

Now if $f(x, y) = c$, it follows from (1) that

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0. \quad (2)$$

In other words, given a one-parameter family of curves $f(x, y) = c$, we can generate a first-order differential equation by computing the differential. For example, if $x^2 - 5xy + y^3 = c$, then (2) gives

$$(2x - 5y)dx + (-5x + 3y^2)dy = 0. \quad (3)$$

For our purposes it is more important to turn the problem around; namely, given a first-order DE such as (3), can we recognize that it is equivalent to the differential $d(x^2 - 5xy + y^3) = 0$?

Definition 2.4.1 Exact Equation

A differential expression $M(x, y)dx + N(x, y)dy$ is an **exact differential** in a region R of the xy -plane if it corresponds to the differential of some function $f(x, y)$. A first-order differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be an **exact equation** if the expression on the left side is an exact differential.

For example, the equation $x^2y^3dx + x^3y^2dy = 0$ is exact, because the left side is $d(\frac{1}{3}x^3y^3) = x^2y^3dx + x^3y^2dy$. Notice that if $M(x, y) = x^2y^3$ and $N(x, y) = x^3y^2$, then $\partial M/\partial y = 3x^2y^2 = \partial N/\partial x$. Theorem 2.4.1 shows that the equality of these partial derivatives is no coincidence.

Theorem 2.4.1 Criterion for an Exact Differential

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a rectangular region R defined by $a < x < b$, $c < y < d$. Then a necessary and sufficient condition that $M(x, y)dx + N(x, y)dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \quad (4)$$

PROOF: (Proof of the Necessity) For simplicity let us assume that $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives for all (x, y) . Now if the expression $M(x, y)dx + N(x, y)dy$ is exact, there exists some function f such that for all x in R ,

$$M(x, y)dx + N(x, y)dy = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy.$$

Therefore, $M(x, y) = \frac{\partial f}{\partial x}$, $N(x, y) = \frac{\partial f}{\partial y}$,

and
$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial N}{\partial x}.$$

The equality of the mixed partials is a consequence of the continuity of the first partial derivatives of $M(x, y)$ and $N(x, y)$. ≡

The sufficiency part of Theorem 2.4.1 consists of showing that there exists a function f for which $\partial f / \partial x = M(x, y)$ and $\partial f / \partial y = N(x, y)$ whenever (4) holds. The construction of the function f actually reflects a basic procedure for solving exact equations.

□ **Method of Solution** Given an equation of the form $M(x, y) dx + N(x, y) dy = 0$, determine whether the equality in (4) holds. If it does, then there exists a function f for which

$$\frac{\partial f}{\partial x} = M(x, y).$$

We can find f by integrating $M(x, y)$ with respect to x , while holding y constant:

$$f(x, y) = \int M(x, y) dx + g(y), \quad (5)$$

where the arbitrary function $g(y)$ is the “constant” of integration. Now differentiate (5) with respect to y and assume $\partial f / \partial y = N(x, y)$:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int M(x, y) dx + g'(y) = N(x, y).$$

This gives
$$g'(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx. \quad (6)$$

Finally, integrate (6) with respect to y and substitute the result in (5). The implicit solution of the equation is $f(x, y) = c$.

Some observations are in order. First, it is important to realize that the expression $N(x, y) - (\partial / \partial y) \int M(x, y) dx$ in (6) is independent of x , because

$$\frac{\partial}{\partial x} \left[N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx \right] = \frac{\partial N}{\partial x} - \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} \int M(x, y) dx \right) = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0.$$

Second, we could just as well start the foregoing procedure with the assumption that $\partial f / \partial y = N(x, y)$. After integrating N with respect to y and then differentiating that result, we would find the analogues of (5) and (6) to be, respectively,

$$f(x, y) = \int N(x, y) dy + h(x) \quad \text{and} \quad h'(x) = M(x, y) - \frac{\partial}{\partial x} \int N(x, y) dy.$$

If you find that integration of $\partial f / \partial x = M(x, y)$ with respect to x is difficult, then try integrating $\partial f / \partial y = N(x, y)$ with respect to y . In either case *none of these formulas should be memorized.*

EXAMPLE 1 Solving an Exact DE

Solve $2xy dx + (x^2 - 1) dy = 0$.

SOLUTION With $M(x, y) = 2xy$ and $N(x, y) = x^2 - 1$ we have

$$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}.$$

Thus the equation is exact, and so, by Theorem 2.4.1, there exists a function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = 2xy \quad \text{and} \quad \frac{\partial f}{\partial y} = x^2 - 1.$$

From the first of these equations we obtain, after integrating,

$$f(x, y) = x^2y + g(y).$$

Taking the partial derivative of the last expression with respect to y and setting the result equal to $N(x, y)$ gives

$$\frac{\partial f}{\partial y} = x^2 + g'(y) = x^2 - 1. \leftarrow N(x, y)$$

It follows that $g'(y) = -1$ and $g(y) = -y$.

Hence, $f(x, y) = x^2y - y$, and so the solution of the equation in implicit form is $x^2y - y = c$. The explicit form of the solution is easily seen to be $y = c/(1 - x^2)$ and is defined on any interval not containing either $x = 1$ or $x = -1$. $\equiv$

Note the form of the solution. It is $f(x, y) = c$.

The solution of the DE in Example 1 is *not* $f(x, y) = x^2y - y$. Rather it is $f(x, y) = c$; or if a constant is used in the integration of $g'(y)$, we can then write the solution as $f(x, y) = 0$. Note, too, that the equation could be solved by separation of variables.

EXAMPLE 2 Solving an Exact DE

Solve $(e^{2y} - y \cos xy) dx + (2xe^{2y} - x \cos xy + 2y) dy = 0$.

SOLUTION The equation is exact because

$$\frac{\partial M}{\partial y} = 2e^{2y} + xy \sin xy - \cos xy = \frac{\partial N}{\partial x}.$$

Hence a function $f(x, y)$ exists for which

$$M(x, y) = \frac{\partial f}{\partial x} \quad \text{and} \quad N(x, y) = \frac{\partial f}{\partial y}.$$

Now for variety we shall start with the assumption that $\partial f / \partial y = N(x, y)$;

that is,
$$\frac{\partial f}{\partial y} = 2xe^{2y} - x \cos xy + 2y$$

$$f(x, y) = 2x \int e^{2y} dy - x \int \cos xy dy + 2 \int y dy + h(x)$$

Remember, the reason x can come out in front of the symbol $\int$ is that in the integration with respect to y , x is treated as an ordinary constant. It follows that

$$f(x, y) = xe^{2y} - \sin xy + y^2 + h(x)$$

$$\frac{\partial f}{\partial x} = e^{2y} - y \cos xy + h'(x) = e^{2y} - y \cos xy \leftarrow M(x, y)$$

and so $h'(x) = 0$ or $h(x) = c$. Hence a family of solutions is

$$xe^{2y} - \sin xy + y^2 + c = 0. \quad \equiv$$

EXAMPLE 3 An Initial-Value Problem

Solve the initial-value problem $\frac{dy}{dx} = \frac{xy^2 - \cos x \sin x}{y(1 - x^2)}, y(0) = 2$.

SOLUTION By writing the differential equation in the form

$$(\cos x \sin x - xy^2) dx + y(1 - x^2) dy = 0$$

we recognize that the equation is exact because

$$\frac{\partial M}{\partial y} = -2xy = \frac{\partial N}{\partial x}.$$

$$\begin{aligned}\text{Now} \quad \frac{\partial f}{\partial y} &= y(1 - x^2) \\ f(x, y) &= \frac{y^2}{2}(1 - x^2) + h(x) \\ \frac{\partial f}{\partial x} &= -xy^2 + h'(x) = \cos x \sin x - xy^2.\end{aligned}$$

The last equation implies that $h'(x) = \cos x \sin x$. Integrating gives

$$h(x) = -\int (\cos x)(-\sin x \, dx) = -\frac{1}{2}\cos^2 x.$$

$$\text{Thus} \quad \frac{y^2}{2}(1 - x^2) - \frac{1}{2}\cos^2 x = c_1 \quad \text{or} \quad y^2(1 - x^2) - \cos^2 x = c, \quad (7)$$

where $2c_1$ has been replaced by c . The initial condition $y = 2$ when $x = 0$ demands that $4(1) - \cos^2(0) = c$ and so $c = 3$. An implicit solution of the problem is then $y^2(1 - x^2) - \cos^2 x = 3$.

The solution curve of the IVP is part of an interesting family of curves and is the curve drawn in blue in **FIGURE 2.4.1**. The graphs of the members of the one-parameter family of solutions given in (7) can be obtained in several ways, two of which are using software to graph level curves as discussed in the last section, or using a graphing utility and carefully graphing the explicit functions obtained for various values of c by solving $y^2 = (c + \cos^2 x)/(1 - x^2)$ for y . $\equiv$

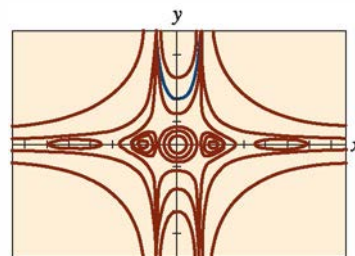


FIGURE 2.4.1 Some solution curves in the family (7) of Example 3

Integrating Factors Recall from the last section that the left-hand side of the linear equation $y' + P(x)y = f(x)$ can be transformed into a derivative when we multiply the equation by an integrating factor. The same basic idea sometimes works for a nonexact differential equation $M(x, y)dx + N(x, y)dy = 0$. That is, it is sometimes possible to find an **integrating factor** $\mu(x, y)$ so that after multiplying, the left-hand side of

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0 \quad (8)$$

is an exact differential. In an attempt to find μ we turn to the criterion (4) for exactness. Equation (8) is exact if and only if $(\mu M)_y = (\mu N)_x$, where the subscripts denote partial derivatives. By the Product Rule of differentiation the last equation is the same as $\mu M_y + \mu_y M = \mu N_x + \mu_x N$ or

$$\mu_x N - \mu_y M = (M_y - N_x)\mu. \quad (9)$$

Although M, N, M_y, N_x are known functions of x and y , the difficulty here in determining the unknown $\mu(x, y)$ from (9) is that we must solve a partial differential equation. Since we are not prepared to do that we make a simplifying assumption. Suppose μ is a function of one variable; say that μ depends only upon x . In this case $\mu_x = d\mu/dx$ and (9) can be written as

$$\frac{d\mu}{dx} = \frac{M_y - N_x}{N} \mu. \quad (10)$$

We are still at an impasse if the quotient $(M_y - N_x)/N$ depends upon both x and y . However, if after all obvious algebraic simplifications are made the quotient $(M_y - N_x)/N$ turns out to depend solely on the variable x then (10) is a first-order ordinary differential equation. We can finally determine μ because (10) is *separable* as well as *linear*. It follows from either Section 2.2 or Section 2.3 that $\mu(x) = e^{\int (M_y - N_x)/N \, dx}$. In like manner it follows from (9) that if μ depends only on the variable y , then

$$\frac{d\mu}{dy} = \frac{N_x - M_y}{M} \mu. \quad (11)$$

In this case, if $(N_x - M_y)/M$ is a function of y only then we can solve (11) for μ .

We summarize the results for the differential equation

$$M(x, y)dx + N(x, y)dy = 0. \quad (12)$$

- If $(M_y - N_x)/N$ is a function of x alone, then an integrating factor for equation (11) is

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}. \quad (13)$$

- If $(N_x - M_y)/M$ is a function of y alone, then an integrating factor for equation (11) is

$$\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}. \quad (14)$$

EXAMPLE 4 A Nonexact DE Made Exact

The nonlinear first-order differential equation $xy \, dx + (2x^2 + 3y^2 - 20) \, dy = 0$ is not exact. With the identifications $M = xy$, $N = 2x^2 + 3y^2 - 20$ we find the partial derivatives $M_y = x$ and $N_x = 4x$. The first quotient from (13) gets us nowhere since

$$\frac{M_y - N_x}{N} = \frac{x - 4x}{2x^2 + 3y^2 - 20} = \frac{-3x}{2x^2 + 3y^2 - 20}$$

depends on x and y . However (14) yields a quotient that depends only on y :

$$\frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3x}{xy} = \frac{3}{y}.$$

The integrating factor is then $e^{\int 3 \, dy/y} = e^{3 \ln y} = e^{\ln y^3} = y^3$. After multiplying the given DE by $\mu(y) = y^3$ the resulting equation is

$$xy^4 \, dx + (2x^2y^3 + 3y^5 - 20y^3) \, dy = 0.$$

You should verify that the last equation is now exact as well as show, using the method of this section, that a family of solutions is $\frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = c$. $\equiv$

Remarks

(i) When testing an equation for exactness, make sure it is of the precise form $M(x, y) \, dx + N(x, y) \, dy = 0$. Sometimes a differential equation is written $G(x, y) \, dx = H(x, y) \, dy$. In this case, first rewrite it as $G(x, y) \, dx - H(x, y) \, dy = 0$, and then identify $M(x, y) = G(x, y)$ and $N(x, y) = -H(x, y)$ before using (4).

(ii) In some texts on differential equations the study of exact equations precedes that of linear DEs. If this were so, the method for finding integrating factors just discussed can be used to derive an integrating factor for $y' + P(x)y = f(x)$. By rewriting the last equation in the differential form $(P(x)y - f(x)) \, dx + dy = 0$ we see that

$$\frac{M_y - N_x}{N} = P(x).$$

From (13) we arrive at the already familiar integrating factor $e^{\int P(x) \, dx}$ used in Section 2.3.

2.4 Exercises

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–20, determine whether the given differential equation is exact. If it is exact, solve it.

- $(2x - 1) \, dx + (3y + 7) \, dy = 0$
- $(2x + y) \, dx - (x + 6y) \, dy = 0$
- $(5x + 4y) \, dx + (4x - 8y^3) \, dy = 0$

- $(\sin y - y \sin x) \, dx + (\cos x + x \cos y - y) \, dy = 0$
- $(2xy^2 - 3) \, dx + (2x^2y + 4) \, dy = 0$
- $\left(2y - \frac{1}{x} + \cos 3x\right) \frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$
- $(x^2 - y^2) \, dx + (x^2 - 2xy) \, dy = 0$